

Case Studies of Remarkable VR Motion Sickness Resistance: A Judo Practitioner and a Commuter

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ABSTRACT

Among 41 participants, two males exhibited no motion sickness (MS) at all during a VR-based cognitive discrimination task. The first, a judo practitioner with 16 years of mid-air balance control training, and the second, a commuter accustomed to reading while commuting, showed suppressed beta activity between cognitive and sensorimotor domains in our EEG analysis, associated with better cognitive behavioral performance. The judo practitioner demonstrated stronger suppression and superior cognitive behavioral performance. These findings raise whether beta activity suppression between cognitive and sensorimotor cortical domains reflects compensatory reticular neuron activation in the deep brainstem under a proposed MS habituation mechanism.

Index terms: Judo, motion sickness, habituation, EEG, cognitive behavioural measurement.

1 INTRODUCTION

According to the visual and vestibular sensory conflict theory, the mismatch between visual and vestibular stimuli is the primary trigger of motion sickness (MS). However, in our ongoing research, we found that 2 out of 41 participants showed no MS at all under this trigger. Based on our interviews with them (see Table 1; a full version of the interview questions can be found in *S. Table 1* in the supplementary document), we identified habituation as a potential explanation. Habituation is one of the most effective therapies for MS [1]. This knowledge has long been widely used in pilot and astronaut training as well as in more recent studies in VR-induced MS [2]. However, the specific brain activity pattern underlying their remarkable resistance to sensory conflict remains unclear. Thus, we conducted case studies on the two participants to explore their anti-MS brain activity patterns. At the same time, since neither case reported experiencing MS, we evaluated whether the two cases demonstrated better cognitive behavioural performance. Thus, this study combines exploratory and hypothesis-confirmatory approaches.

2 RELATED WORK

Our ongoing research induces MS by employing a cognitive discrimination ability assessment paradigm described in [3]. This paradigm involves creating sensory conflict through a 30-minute exposure to mismatched vestibular stimulus (rotational motion from a rotating chair-based setup) and visual stimulus (VR-based forward-moving visual motion). A total of 41 participants (21 male, 20 female) were recruited. To ensure consistent vestibular and visual stimulation, participants were instructed to avoid head

movements throughout the experiment. All participants completed the 30-minute cognitive task, with 39 reporting symptoms of MS.

Table 1. Interview Questions and Responses

Questions	Case I	Case II
Are you playing VR games a lot?	Never, apart from your experiment	No, 4 times in my life
Are you playing video games a lot?	No, since I was roughly 14, I haven't owned any video game console.	No, just a little bit in high school, maybe 4 hours a week
Are you a professional or amateur sports player?	Yes, since I was 8, I have been practicing judo. So, now it has been 16 years. I have been thrown in the air, trying to regain balance mid-air or as soon as I land on the mat. (I would say that this might be the greatest contributor to motion-sickness resistance)	No.
Other clues that you want to share	As I mentioned, I believe that practicing judo is the main contribution to motion-sickness resistance. Judo requires a great deal of spatial awareness and being in control of your body at any time.	I used to reading on the bus, in a car etc., so maybe I trained myself that way?

3 METHODS

We used resting-state EEG signals *before* and *after* the cognitive task to minimize cognitive "noise" and isolate habituation-induced EEG patterns. We also examined response time (RT) *during* the task to assess whether remarkable MS resistance improved cognitive performance. Seven-channel EEG data were collected using the StarStim 8 system (Neuroelectronics, Spain), targeting the newly proposed human vestibular network (HVN) [4] across cognitive (Fz, Pz) and sensorimotor (P3, P4, CP5, CP6) domains. EEG and RT data from the 39 participants who reported MS were grand averaged as the baseline for EEG and RT analysis.

3.1 EEG analysis

Brain functional connectivity (BFC) analysis between cognitive and sensorimotor domain based on phase-locking values (PLVs) of inter-electrode coherence (IEC) [5] at delta (δ : 0.5 to 3Hz), theta

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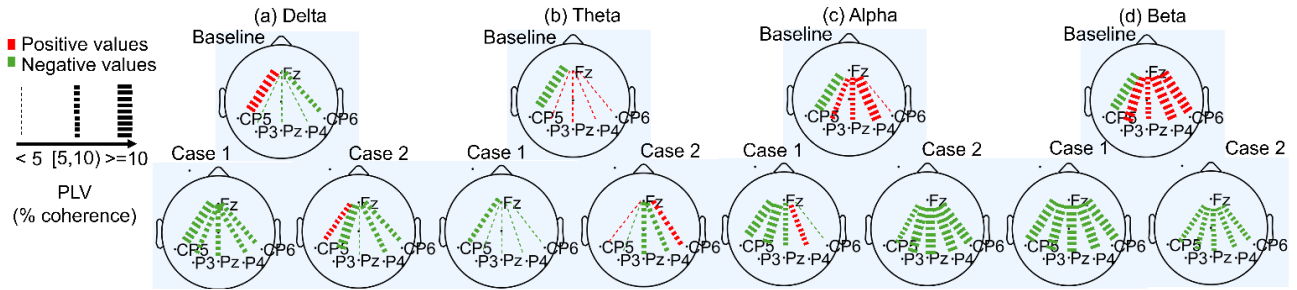


Figure 1: Comparisons of brain activity patterns between anti-MS individuals and baseline conditions

(θ :4-7Hz), alpha (α :8-12Hz) and beta (β :13-20Hz) frequency bands was conducted for EEG data. More details are as follows:

1. BFC (Fz to P3): frontal midline in cognitive domain to left parietal insular vestibular cortex (PIVC) in sensorimotor domain
2. BFC (Fz to P4): frontal midline to right PIVC
3. BFC (Fz to CP5): frontal midline to left temporoparietal Junction (TPJ) in sensorimotor domain.
4. BFC (Fz to CP6): frontal midline to right TPJ
5. BFC (Fz to Pz): frontal midline to posterior midline within the cognitive domain

The 5 electrode pairs with 4 frequency bands generated a matrix of 20 EEG features. Ultimately, each feature in “after” condition was corrected by its “before” condition using the equation below:

$$\text{Corrected feature} = \frac{\text{After} - \text{Before}}{\text{Before}} \times 100\% \quad (1)$$

3.2 Behavioral analysis

RT, a widely used indicator of cognitive discrimination ability was used to evaluate how fast a participant can accurately discriminate between targets (a green circle) and non-targets (other circles) in the cognitive task. Details about this MS induction task are in [3].

4 RESULTS

As shown in Fig. 1(a)-(d) and *S. Table 2* (see the supplementary document), both cases presented decreased post-MS BFC across all frequency bands between Fz and P3, as well as between Fz and Pz. This indicates suppressed BFC between the frontal midline in the cognitive domain and the left PIVC in the sensorimotor domain, as well as between the frontal midline and the posterior midline within the cognitive domain. Furthermore, the two cases showed the greatest similarity in the beta frequency band, characterized by suppression across all five BFC directions. However, Case 1 (judo practitioner) exhibited stronger suppression than Case 2 (Commuter reading on the move), with values exceeding 10% compared to the 5%-10% range observed in Case 2. We also confirmed that both cases exhibited better cognitive behavioural performance based on their RTs, with Case 1 (482.709ms) showing faster RTs than Case 2 (517.461ms) and baseline (594.245ms). That is, Case 1 and Case 2 were faster than the baseline by over 100ms and 75ms, respectively. In summary, the judo practitioner demonstrated more pronounced pre-post MS BFC changes, which were associated with superior cognitive behavioural performance.

5 DISCUSSION

It has been vaguely proposed that the compensatory activation of reticular neurons in the brainstem in response to specific motion stimuli, which allows resistance to vestibular-cerebellar input and maintains normal cortical arousal, may serve as the mechanism

underlying habituation to MS [6]. Our findings prompt the question of whether the suppression of beta activity between the cognitive and sensorimotor domains within the HVN represents a compensatory activation of reticular neurons at the cortical level, detectable through EEG. This is particularly relevant given a recent study demonstrating the effectiveness of cybersickness reduction through a neuromodulation technique that disrupts beta activity in the left PIVC (represented by P3) [7]. Furthermore, we can confidently assert that habituation facilitates a far transfer effect, extending resistance from specific motion to non-specific motion, as judo training appears unrelated to our MS induction task. Additionally, we can rule out the possibilities that heightened attention to the cognitive task and better postural stability reduced the two individuals’ MS [8][9], as some participants who experienced MS exhibited even better cognitive performance and more stable posture (see supplementary document). In future research, we plan to recruit more fully MS-habituated participants to further explore EEG-detectable habituation mechanism.

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